

Hoang Minh Thanh

Machine Learning Engineer

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I focus on full-stack development of TTS, ASR, LLMs, from deep learning research knowledge to fine-tuning and production deployment. Adapting state-of-the-art open-source models using parameter-efficient methods (LoRA, QLoRA). I bring a combination of fundamental research depth and engineering execution.

★ Keypoint

- **Solid fundamental knowledge of Deep Learning:** Proven by my [thesis report](#), creator of [SurveyWithCode](#), proposal [DeepGesture](#). Experience in building evaluation pipelines for [perplexity](#), toxicity, and relevance to ensure safe and high-quality model outputs.
- **Experience in LLM Fine-Tuning & Distill & LoRA, QLoRA:** Proven by fine-tune F5-TTS model to create [Par-rotSpeech](#) TTS in Vietnamese, develop Chat LLM [MillionScope](#). Experience in SGLang, export ONNX, TensorRT, deploying models in production to prevent drift and ensure high-performance, reliable service.
- **Skilled in Prompt Engineering & Data Curation:** Adept at engineering prompts and curating datasets that reflect precise tone, brand voice, and safety guidelines, ensuring model outputs are both trustworthy and aligned with product objectives.
- **Problem solving and modern AI optimization technique:** FlashAttention, Attention Sink, KV Cache, CUDA Kernel C++

🎓 Education

AS	University of Science - VNUHCM, Information Technology	2012 – 2015
BS	University of Science - VNUHCM, Computer Science	Sept. 2018 – Sept. 2020
	• GPA: 3.3/4.0 • Thesis: GCAT - Link Prediction in Knowledge Graph, [Report], [Code]	
MSc	University of Science - VNUHCM, Computer Science	Oct. 2021 – Dec. 2024
	• GPA: 3.44/4.0 • Thesis: OpenHuman: A conversational gesture synthesis system based on emotions and semantics, [Report], [Code]	

📖 Publications

DeepGesture: A conversational gesture synthesis system based on emotions and semantics [arXiv] , [homepage]	Jul. 2025
Towards a GENE Leaderboard [arXiv]	Oct. 2024
Graph Collaborative Attention Network for Link Prediction in Knowledge Graphs [arXiv] [homepage]	Oct. 2020

💼 Experience

Rakumo Inc, Software Engineer	Jul. 2019 – Apr. 2020
• Developed a Django-based web application with integrated DocuSign.	
FPT Software, AWS Data Engineer	Oct. 2020 – Nov. 2021
• CI/CD pipeline terraform in Jenkins, implemented and modified real-time and batch data pipelines on AWS using Kinesis, Lambda, S3, Glue, StepFunction,	

SNS, ..etc..

- Cox Automotive: Collaborated with international teams in Agile

VNG Corp, Software Engineer

Oct. 2021 – May 2024

ZDN Team (Zalo Content Delivery Network): caching service serve millions request

- Implement scheduled `ScheduledThreadPool` for monitoring stats execution.
- Built admin dashboard with server visualization via `amchart`, `p5.js` in `NextJS`.
- Upgrade internal Golang package for Apache Thrift

OpenHuman (openhuman.ai), Creator

May 2024 – Present

- Join GENE Research Team ([GENEA Leaderboard](#), [HEMVIP](#)): Designed and built a full-stack, production-ready leaderboard system to evaluate multiple AI gesture generation models, integrating user study workflows via Prolific.

</> Projects

MillionScope (End-to-end ownership: [Code](#))

millionscope.com

- Finetune GPT-OSS 20B via OpenAI-Cookbook with custom dataset.
- Experiment post-training RLHF chat response via [huggingface/trl](#), evaluate perplexity with [huggingface/evaluate](#)
- Develop full-stack both UI (Nextjs, Chat SDK, Cloudflare Pages) and backend LLM serving with SGLang on Thunder Compute, Cloudflare AI

ParrotSpeech (End-to-end ownership: [Code](#))

parrotspeech.pages.dev

- Crawl speech dataset with `yt-dlp` youtube channels. Using `Chunkformer` to transcribe, preprocessing dataset.
- Fine-tune F5-TTS with Vietnamese speech and quantization QLoRA with `bitsandbytes`
- Export weights and serving via `onnx-runtime` and `transformers.js`

DeepGesture ([Code](#)): Convert euler rotation motion to quaternion-space, develop Diffusion Transformer (DiT) for generate MFCC coefficient (Similar MFCC speech but for skeleton motion movement).

deepgesture.pages.dev

SurveyWithCode ([Code](#)): PapersWithCode alternative using D3.js visualize taxonomy survey, provide DeepLearning knowledge.

surveywithcode.pages.dev

Others pet projects: [Realisitic Digital Human Render Engine](#), [Game Pucca Runner](#): dijkstra algorithm path finding game; [3D Threejs Skeleton](#), [MetaPet](#) ([metapet.vercel.app/Code](#)), [cortexpod.com](#); [pithagon.com](#)

🌟 Certifications

- [Critical Thinking Certification 2014](#), [Emotion Recognition Certification 2019](#), [Summer ML](#)
- Coursera: [Design Pattern](#), [TensorFlow Developer Professional](#) 2021
- **AWS:** [Solutions Architect - Associate](#), [Machine Learning - Specialty](#); [Solution Architect](#) 2021

💻 Technologies

Programming: Python, Javascript, C++, CUDA Kernel, SQL, Bash

Frameworks & Libraries: PyTorch, SGLang, `transformers.js`, PEFT, TRL, `bitsandbytes`, [confident-ai/DeepEval](#)

Fine-Tuning: QLoRA, SFT, RLHF, Knowledge Distillation

Evaluation: [perplexity](#), BLEU, ROUGE, Custom Metric Design (Toxicity, Relevance, Brand Voice)

MLOps & Deployment: Docker, Kubernetes, AWS SageMaker, `wandb`, MLflow, Grafana